ATOPARY DEVELOPERS AND REALTY

A T O P A R Y P R O P E R T I E S

DIGITAL PLATFORM

PRODUCT SCOPE OF WORK

Website Upgrade · Rental Management Portal · Payment & Booking Engine

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1. EXECUTIVE SUMMARY

Atopary Developers and Realty is committed to building a world-class digital real estate platform that reflects the quality, ambition, and innovation at the heart of the Atopary brand. This Scope of Work document outlines the full product vision, feature requirements, and technical specifications for the next phase of the Atopary digital platform — a comprehensive upgrade that will elevate the website into a fully functional, high-performance property sales and rental ecosystem.

This is not an exercise in replication. The platform to be built is an original Atopary product, designed from the ground up to serve the unique needs of Atopary's clients, investors, and property partners across Ghana and beyond. Every feature, user journey, and interface element is to be developed in alignment with Atopary's brand identity and strategic business objectives.

The platform encompasses two core pillars:

- The Atopary Sales Portal — a premium, dynamic, and highly engaging digital front for property discovery, listings, agent interaction, and lead generation;
- The Atopary Rental Hub — a full-featured rental property marketplace and management system, enabling property owners to list, manage, and monetise short-stay and long-stay rental properties through Atopary's platform.

2. PROJECT OBJECTIVES

The following core objectives define the success criteria for this engagement:

01	Deliver a fast, modern, mobile-first website experience that reflects the premium quality of the Atopary brand.
02	Build a dynamic and fully searchable property listings system with rich media, advanced filters, and real-time updates.
03	Create an intuitive user journey from property discovery through to enquiry, appointment, and deal initiation.
04	Develop a standalone Atopary Rental Hub — a proprietary rental marketplace allowing property owners to self-list, manage bookings, and receive payments end-to-end.
05	Integrate a secure, multi-currency payment gateway supporting local and international transactions.
06	Implement a fully automated booking and calendar management system for rental properties.
07	Establish a robust backend CMS enabling the Atopary team to manage all content, listings, and user accounts independently.
08	Ensure the platform is SEO-optimised, analytics-ready, and built for long-term scalability.

3. PILLAR ONE — ATOPARY SALES PORTAL UPGRADE

The Atopary Sales Portal is the primary digital face of the business and must function as a premium, highly engaging platform that inspires confidence in buyers and investors. The following features are required across the upgraded portal.

3.1 Homepage & Global Navigation

- A visually impactful, full-screen hero section with high-resolution property imagery, dynamic headline text, and a prominent property search bar;
- Instant search functionality accessible directly from the homepage — allowing users to search by location, property type, price range, and number of bedrooms;
- A clean, sticky navigation bar with clear sections: Buy, Rent, New Developments, About, Contact, Agent Login, and User Account;
- A curated "Featured Listings" section showcasing selected properties with key details, pricing, and a direct call-to-action;
- A dynamic "New on the Market" section that auto-updates as new properties are added to the platform;
- A statistics bar displaying key brand metrics (e.g. properties sold, years of experience, client satisfaction, active listings) with animated counters;
- Testimonials carousel featuring client reviews with photos and property details;
- A "Latest News & Insights" blog section for market updates, property guides, and investment tips;
- Clear CTAs throughout — "Schedule a Viewing", "Get a Free Valuation", "Speak to an Agent";
- Full mobile responsiveness across all screen sizes with optimised load performance.

3.2 Property Listings & Search

- A fully searchable, filterable property listings page with grid and map view toggle;
- Advanced search filters: location, property type, price range (min/max), bedrooms, bathrooms, plot size, furnishing status, year built, and amenities;
- Interactive map view (Google Maps / Mapbox integration) with clickable property pins that display listing previews;
- Each listing card to display: primary photo, price, location, bedroom and bathroom count, property type badge, and "Save" / "Compare" actions;
- Sort options: Newest First, Price Low–High, Price High–Low, Most Viewed;
- Pagination with configurable listings per page and infinite scroll option;
- "Save Search" functionality that notifies users when new matching listings are added;
- Listing comparison tool allowing users to compare up to three properties side by side.

3.3 Individual Property Detail Pages

- Full-screen, high-resolution photo gallery with lightbox zoom and slideshow functionality;
- Embedded 3D virtual tour or 360° photo viewer where applicable;
- Embedded video walkthrough capability (YouTube / Vimeo / direct upload);
- Comprehensive property details: description, specifications, floor plan, amenities checklist, and neighbourhood information;
- Interactive map showing the property location, nearby schools, hospitals, transport, and commercial centres;

- Dynamic pricing display with currency toggle (GHS / USD);
- "Schedule a Viewing" form integrated into the listing page;
- "Make an Enquiry" direct chat or form module;
- Agent profile card with photo, credentials, and direct contact options;
- "Similar Properties" recommendation section at the bottom of each listing;
- Social media share buttons and "Copy Link" functionality;
- Print-friendly property brochure download (auto-generated PDF).

3.4 User Accounts & Dashboard

- User registration and login via email, Google OAuth, or phone number OTP;
- Personal dashboard showing: Saved Listings, Search Alerts, Viewing Appointments, Enquiry History, and Documents;
- Wishlist / saved properties with custom notes and status tags;
- Viewing appointment scheduler with calendar integration and automated reminders via email and SMS;
- Document vault for storing purchase-related documents (agreements, receipts, correspondence) securely;
- Profile management: photo, contact details, preferences, and notification settings.

3.5 Agent & Team Portal

- Individual agent profile pages with bio, photo, listings, and performance stats;
- Agent CRM dashboard: leads received, viewings scheduled, enquiries open, and deals in progress;
- Lead assignment system routing enquiries to the correct agent automatically;
- Commission tracking module linked to the compensation structure;
- Internal messaging and notification system between agents and management.

3.6 Content, Media & SEO

- A fully managed blog / insights section with categories: Market Updates, Investment Guides, Area Spotlights, Legal & Finance;
- SEO-optimised page structures for all listing pages, blog posts, and key landing pages;
- Auto-generated meta titles, descriptions, and schema markup for property listings;
- Image optimisation and lazy loading for fast page performance;
- Google Analytics 4, Meta Pixel, and Google Tag Manager integration;
- XML sitemap and robots.txt configuration;
- Open Graph and Twitter Card meta tags for rich social media previews.

3.7 Performance & Technical Standards

- Target page load time: under 2.5 seconds on mobile and desktop;
- Google Lighthouse performance score target: 90+ across all categories;
- HTTPS with SSL certificate and robust security protocols;
- WCAG 2.1 accessibility compliance;
- Cross-browser compatibility: Chrome, Safari, Firefox, Edge;

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- Progressive Web App (PWA) capability for mobile users;
- CDN integration for fast global content delivery.

4. PILLAR TWO — ATOPARY RENTAL HUB

The Atopary Rental Hub is a proprietary rental property marketplace — a standalone but integrated module within the Atopary platform that enables property owners to self-list their rental properties and allows guests and tenants to discover, book, and pay for accommodation directly through Atopary. This module positions Atopary as the definitive digital home for property rentals in Ghana, operating as a branded platform with Atopary's standards, trust frameworks, and service quality embedded throughout.

4.1 Rental Hub — Host Registration & Onboarding

Any property owner wishing to list a rental on the Atopary Rental Hub must complete a structured onboarding process:

- Host registration: sign up with full name, contact details, NIN / passport ID, and bank account details for payouts;
- Identity verification: document upload and optional video selfie verification step to ensure platform trust and security;
- Host profile creation: photo, bio, hosting experience, response rate, and languages spoken;
- Atopary Host Agreement acceptance — a digital terms-of-service covering platform standards, cancellation policies, and commission structure;
- Onboarding wizard guiding the host through each step with progress tracking and contextual help text;
- Host verification badge awarded upon successful completion of identity checks.

4.2 Property Listing Creation

Hosts can create listings through a seamless, guided multi-step listing builder:

- Step 1 — Property Type: apartment, house, villa, studio, guesthouse, commercial space, boys' quarters, and more;
- Step 2 — Location: GPS-enabled address input with map pin placement for precision;
- Step 3 — Specifications: number of bedrooms, bathrooms, maximum guests, floor area, and accessibility features;
- Step 4 — Photos & Media: drag-and-drop photo uploader (minimum 5 images required), video upload, and optional 360° virtual tour embed;
- Step 5 — Title & Description: AI-assisted description tool to help hosts write compelling, professional listing copy;
- Step 6 — Amenities: categorised checklist (Wi-Fi, Air Conditioning, Pool, Generator, Security, Kitchen, Parking, Laundry, etc.);
- Step 7 — House Rules: pet policy, smoking policy, noise curfew, visitor policy, check-in instructions;
- Step 8 — Pricing: nightly rate, weekly discount, monthly discount, cleaning fee, security deposit, and minimum stay;
- Step 9 — Availability: calendar-based availability management — block dates, set advance notice periods, and define booking windows;

- Step 10 — Review & Publish: full listing preview before going live, with edit capability at any step.

4.3 Booking & Availability Management

- Real-time availability calendar embedded on each listing — guests can see available dates at a glance;
- Instant Book mode (for qualified hosts): guests can book without prior host approval;
- Request to Book mode: host reviews and approves each booking request within a defined window;
- Automated booking confirmations sent to both guest and host via email and in-app notification;
- Booking request expiry: unresponded requests auto-expire and dates are released back to the calendar;
- Multi-calendar management: hosts managing multiple properties can view all bookings across all listings on a single unified calendar dashboard;
- iCal sync: hosts can connect external calendars (Google Calendar, iCloud) to automatically block dates from other platforms;
- Automated pre-arrival messaging sent to guests with check-in details a configurable number of days before arrival;
- Automated post-checkout review requests sent to guests.

4.4 Payment Gateway & Financial Management

A fully integrated, multi-channel payment system shall be built into the Atopary Rental Hub, supporting both local and international transactions:

- Supported payment methods: mobile money (MTN MoMo, Vodafone Cash, AirtelTigo), Ghana bank debit/credit cards, international Visa/Mastercard, and PayPal;
- Multi-currency support: GHS (Ghana Cedi) and USD with live exchange rate display;
- Payment held in escrow upon booking and released to the host 24 hours after guest check-in, protecting both parties;
- Atopary platform commission automatically deducted at source before host payout is processed;
- Automated host payout system: configurable payout schedule (weekly or per-booking) to registered mobile money or bank account;
- Security deposit handling: collected at booking, held digitally, and released or partially withheld after the stay based on host report;
- Automated payment receipts and invoices generated for every transaction for both guest and host;
- Full transaction history accessible in both guest and host dashboards;
- Refund management: automated refund processing based on the applicable cancellation policy;
- PCI-DSS compliant payment infrastructure — all card data handled through a certified payment processor (e.g. Paystack, Flutterwave, or Stripe).

4.5 Guest Experience & Discovery

- Dedicated Rental Hub homepage with destination-based search: location, check-in date, check-out date, number of guests;

- Advanced rental filters: property type, price per night, bedrooms, amenities, instant book, host rating, distance from landmarks;
- Map-based rental search with live price pins and property preview cards;
- Each rental listing page displays: full photo gallery, all amenities, house rules, host profile, availability calendar, and dynamic pricing breakdown;
- Guest registration and login — guests can use their existing Atopary account or register separately;
- Guest dashboard: upcoming bookings, past stays, saved listings, messages, and payment history;
- In-platform guest–host messaging system with read receipts and notification alerts;
- Automated check-in information delivery with access codes, directions, and host notes.

4.6 Reviews & Trust Framework

- Dual review system: guests review properties and hosts; hosts review guests — both submitted after checkout;
- Star ratings across: Cleanliness, Accuracy, Location, Communication, Value, and Overall Experience;
- Review moderation: all reviews subject to Atopary's content guidelines before publication;
- Host response capability: hosts can publicly respond to guest reviews;
- Trust badges: Identity Verified, Superhost (based on performance metrics), Instant Book;
- Host performance dashboard: average rating, total reviews, response rate, acceptance rate, and Superhost status tracker.

4.7 Host Dashboard & Management Tools

- Comprehensive host dashboard overview: total bookings, revenue this month, upcoming check-ins, and pending requests;
- Listing management: edit any listing detail, update pricing, adjust minimum stay, add seasonal rates;
- Dynamic pricing tool: automatic rate suggestions based on local demand, seasonality, and occupancy trends;
- Cancellation policy configuration: Flexible, Moderate, Strict — with Atopary policy parameters applied;
- Payout settings: add and manage multiple payout accounts with priority selection;
- Performance analytics: views per listing, booking conversion rate, average nightly rate, and revenue trends by month;
- Dispute management: structured process for reporting property damage, missing items, or guest conduct issues;
- Tax documentation: downloadable annual income summary for tax reporting purposes.

4.8 Platform Administration — Rental Hub

- Atopary admin panel to manage all hosts, guests, listings, and bookings from a central dashboard;
- Listing approval queue: review and approve or reject new listings before they go live;
- Commission management: configure platform commission percentage by property type or category;

- Dispute resolution centre: admin-mediated dispute process with documented evidence trails;
- User flagging and suspension system for hosts or guests in breach of platform terms;
- Financial reporting: total platform revenue, commissions collected, payouts processed, and outstanding balances by period;
- Promotional tools: create featured listing campaigns, discount codes, and seasonal promotions.

5. CONTENT MANAGEMENT SYSTEM (CMS)

A robust, user-friendly CMS shall be delivered to empower the Atopary team to independently manage all platform content, listings, and users without developer dependency.

- Full property listing CRUD (Create, Read, Update, Delete) with rich media management;
- Blog and news article editor with full formatting, image uploads, category tags, and SEO fields;
- Team and agent profile management;
- Homepage content management: hero banners, featured listings, statistics, testimonials, and promotions;
- Lead and enquiry management: view, assign, and track all incoming leads;
- User account management: view, edit, suspend, or delete user accounts;
- Roles and permissions: Admin, Agent, Content Editor, Finance — each with scoped access;
- Platform-wide notification management: push, email, and SMS announcements.

6. TECHNICAL ARCHITECTURE & STACK RECOMMENDATIONS

The following technology recommendations are provided as a guide. The development team may propose alternatives subject to approval, provided they meet the performance, scalability, and security requirements of this scope.

Layer	Recommended Technology / Approach
Frontend Framework	Next.js (React) — for SSR, SEO, and performance
Styling	Tailwind CSS with Atopary custom design system
Backend / API	Node.js with Express or NestJS; RESTful + GraphQL APIs
Database	PostgreSQL (relational) + Redis (caching)
File & Media Storage	AWS S3 or Cloudinary for image and video management
Authentication	JWT + OAuth2 (Google, Facebook) + OTP (SMS via Twilio/Africa's Talking)
Payment Gateway	Paystack (primary for Ghana) + Stripe (international)
Maps Integration	Google Maps API (property pins, search, nearby amenities)
Search	Algolia or Elasticsearch for fast, filterable property search

Email / Notifications	SendGrid (email) + Firebase Cloud Messaging (push) + Africa's Talking (SMS)
Calendar Sync	iCal / CalDAV protocol for third-party calendar integration
Hosting / Infrastructure	AWS or Google Cloud with auto-scaling, load balancing, and CDN
CI/CD	GitHub Actions or GitLab CI with staging and production environments
Analytics	Google Analytics 4 + Hotjar (user behaviour) + Mixpanel (events)

7. PHASED DELIVERY PLAN

The project is recommended to be delivered in four structured phases to ensure controlled quality, stakeholder review at each milestone, and manageable deployment risk.

Phase 1 — Foundation & Design System — *Weeks 1–4*

- UX/UI design for all pages (desktop + mobile)
- Atopary design system: typography, colours, components
- Homepage, navigation, and core sales portal pages
- CMS infrastructure setup
- Development environment, CI/CD pipeline, and staging server

Phase 2 — Sales Portal — Full Build — *Weeks 5–10*

- Full property listings system with search and filters
- Individual property detail pages with gallery and virtual tour
- User registration, accounts, and dashboard
- Agent profiles and agent CRM
- SEO optimisation and analytics integration
- Blog / insights module

Phase 3 — Rental Hub — Full Build — *Weeks 11–18*

- Host registration and onboarding system
- Property listing wizard (all 10 steps)
- Availability calendar and booking engine
- Payment gateway integration (Paystack + Stripe)
- Guest dashboard and messaging system
- Review and trust framework
- Host analytics and payout management

Phase 4 — Testing, Optimisation & Launch — *Weeks 19–22*

- Full QA testing across all devices and browsers

- Security audit and penetration testing
- Performance optimisation (Lighthouse 90+ target)
- UAT (User Acceptance Testing) with Atopary team
- Soft launch with selected users
- Full public launch and post-launch monitoring

8. FEATURE PRIORITY MATRIX

The table below classifies all major features by delivery priority. Critical features must be present at launch. High priority features are expected in Phase 2 or 3. Standard features may be delivered post-launch in a subsequent sprint.

Feature	Description	Priority
Homepage & Hero Search	Dynamic homepage with full search bar and featured listings	**Critical**
Property Listings Page	Full filterable listings grid with map toggle	**Critical**
Property Detail Pages	Full media gallery, specs, map, enquiry, and agent card	**Critical**
User Registration & Login	Email, Google OAuth, OTP sign-up and login	**Critical**
Mobile Responsiveness	Fully responsive across all devices	**Critical**
CMS Backend	Full content management for listings, blog, and team	**Critical**
Virtual Tours / Video	Embedded 3D/360° and video walkthrough support	**High**
Property Comparison Tool	Side-by-side comparison of up to 3 properties	**High**
Agent CRM Dashboard	Lead management and commission tracking for agents	**High**
Blog / Insights Module	SEO-driven content hub with categories and search	**High**
Host Onboarding	Step-by-step rental host registration and verification	**Critical**
Rental Listing Builder	10-step guided listing creation wizard	**Critical**
Booking Engine	Instant Book and Request to Book with calendar management	**Critical**
Payment Gateway	Paystack + Stripe + Mobile Money integration	**Critical**
Rental Multi-Calendar	Unified calendar dashboard across all host listings	**Critical**
iCal Sync	Sync availability with external calendar platforms	**High**
Guest–Host Messaging	In-platform secure messaging with notifications	**Critical**
Review System	Dual post-stay review system with star ratings	**High**
Host Payout System	Automated payouts to mobile money / bank account	**Critical**
Dynamic Pricing Tool	Demand-based rate suggestions for rental hosts	**Standard**

PWA / Mobile App Ready	Progressive Web App optimisation	**Standard**
Admin Rental Panel	Commission, dispute, and listing moderation tools	**High**

9. NON-FUNCTIONAL REQUIREMENTS

9.1 Security

- All user data encrypted at rest and in transit (TLS 1.2+);
- PCI-DSS compliance for all payment processing flows;
- OWASP Top 10 vulnerability mitigation applied throughout development;
- Two-factor authentication available for all admin and agent accounts;
- GDPR-aligned privacy framework with clear data consent management.

9.2 Scalability & Reliability

- Architecture designed to support up to 100,000 concurrent users without performance degradation;
- Auto-scaling infrastructure on AWS or Google Cloud;
- 99.9% uptime SLA target with monitoring and automated alerts;
- Database backup every 6 hours with 30-day retention;
- Disaster recovery plan with documented RTO and RPO.

9.3 Accessibility & Internationalisation

- WCAG 2.1 Level AA accessibility compliance;
- Multi-language foundation: English primary, with architecture ready for French and Twi localisation;
- Multi-currency display: GHS and USD with live rate conversion.

10. ASSUMPTIONS AND EXCLUSIONS

10.1 Assumptions

- Atopary will provide all brand assets (logo, colours, typography, photography) in production-ready format;
- All property listing content (descriptions, photos, pricing) for the initial launch listings will be provided by Atopary;
- Payment gateway merchant accounts (Paystack, Stripe) will be created and owned by Atopary — the developer will handle integration only;
- Atopary will designate a project lead for approvals, content review, and UAT sign-off at each phase;
- Domain, hosting, and third-party API costs are to be borne by Atopary.

10.2 Exclusions

- Native iOS or Android mobile app development (not included in this scope — PWA covers mobile access);
- Third-party property portal syndication (e.g. Meqasa, Jumia House) — may be considered in a future phase;
- Physical print or offline marketing materials;
- Ongoing post-launch maintenance beyond a mutually agreed defect warranty period (to be specified in the development agreement).

11. KEY DELIVERABLES

- UX/UI Design Files (Figma): all screens, components, mobile and desktop layouts;
- Fully functional and deployed Atopary Sales Portal;
- Fully functional and deployed Atopary Rental Hub;
- CMS backend with admin access and full documentation;
- Integrated payment gateway with test and live mode;
- Automated booking, calendar, and notification system;
- Full technical documentation: API docs, database schema, deployment guide;
- User guide for the Atopary admin and content team;
- 90-day post-launch defect warranty and support window.

12. SCOPE APPROVAL

This Scope of Work document requires formal sign-off from both the Atopary authorised representative and the appointed development team lead before any development work commences. Any changes to scope after sign-off are subject to a formal change request process and may impact timeline and cost.

FOR ATOPARY DEVELOPERS AND REALTY	DEVELOPMENT TEAM REPRESENTATIVE
Signature: _____	Signature: _____
Name: _____	Name: _____
Title: _____	Company: _____
Date: _____	Date: _____

— END OF SCOPE OF WORK —

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